

DN-CBM: Task-Agnostic Concept Bottlenecks via Automated Concept Discovery

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Abstract

Concept Bottleneck Models (CBMs) enhance interpretability by mapping images to human-understandable concepts before classification. However, selecting relevant concepts and aligning feature representations to these concepts remain challenging, even with powerful foundation models like CLIP. We propose Discover-then-Name-CBM (DN-CBM) which inverts the typical approach: instead of pre-selecting concepts, we use sparse autoencoders to first discover inherent model concepts, then automatically name them, and finally train linear probes for classification. This method is task-agnostic, efficient, and leverages existing model knowledge. Comprehensive evaluations show that our approach yields meaningful concepts resulting in interpretable and performant CBMs. Code will be made available.

1. Introduction & Related Work

Deep neural networks are powerful but lack transparency, posing risks in safety-critical applications. Explainable models help detect biases, model failures, improve fairness, and build trust. For example, concept bottleneck models (CBMs) [1, 4, 9–12, 15, 17–19, 27, 29–31, 33] provide interpretability by detecting human-interpretable concepts from model features and linearly combining them to perform classification (Fig. 1c). Recent CBMs leverage large-language models (LLMs) such as GPT-3 [3] to collect task-relevant concepts and vision-language models such as CLIP [21] to learn the mapping from model features to concepts in an attribute-label-free manner [15, 17, 19, 30]. In this process, the model is forced to align to these fixed sets of concepts, and it is not clear if these predefined sets of concepts can be faithfully detected by the model [14, 25].

To address this, we propose DN-CBM which inverts the typical CBM paradigm: instead of defining concepts beforehand, (1) it *discovers* what the model already *knows* (Fig. 1a), (2) names these concepts automatically (Fig. 1b), and then (3) performs classification (Fig. 1c). For the first step, we leverage Sparse Autoencoders (SAEs). Inspired by

their success in the language domain [2, 5], we use them to discover human interpretable concepts known to the model and find them to work well in vision as well. Doing so, our approach eliminates the need to query LLMs for task-relevant concepts. Moreover, we propose a novel method to assign names to the discovered concepts by leveraging the vision-language alignment in CLIP. Finally, once we learn the latent named concept space, we keep it frozen to train classifiers to construct CBMs for a variety of downstream classification tasks, which renders the method task-agnostic and computationally efficient. We find that the discovered concepts are semantically meaningful and the assigned names are appropriate (Fig. 2a), rendering them easy to interpret (Fig. 2b-c), yielding performant (Fig. 2d) and interpretable CBMs (Fig. 2e).

2. Constructing DN-CBM

In this section, we outline our concept discovery and naming method and how we use it to build our DN-CBM.

Discover concepts known to the model (Fig. 1a): To extract concepts, we use a setup similar to [2] and train SAEs on CLIP ViT-B/16 final layer features using the CC3M dataset [28]. The SAE consists of a linear encoder $f(\cdot)$ with weights $\vec{W}_e \in \mathbb{R}^{d \times h}$, a ReLU non-linearity ϕ , and a linear decoder $g(\cdot)$ with weights $\vec{W}_d \in \mathbb{R}^{h \times d}$. For a given input $\vec{a}$, it computes: $\text{SAE}(\vec{a}) = (g \circ \phi \circ f)(\vec{a}) = \vec{W}_d^T \phi(\vec{W}_e^T \vec{a})$. The hidden representation is very high dimensional (i.e. $h \gg d$) and is quite sparse. It is trained with an L_2 reconstruction loss, and an L_1 sparsity regularisation, with λ_1 as a hyperparameter, $\mathcal{L}_{\text{SAE}}(\vec{a}) = \|\text{SAE}(\vec{a}) - \vec{a}\|_2^2 + \lambda_1 \|\phi(f(\vec{a}))\|_1$. After training, the SAE latent features can be interpreted as discovered concepts known to the model.

Name the discovered concepts (Fig. 1b): To name the discovered concepts, we use the set of 20k most frequent English words as used in [16] as the vocabulary $\mathcal{V}$ and get their embeddings $\mathcal{E}$ using the CLIP text encoder. Each of the SAE neurons c is assigned a specific dictionary vector $\vec{w}_c$, corresponding to a column of the decoder weight matrix, $\vec{w}_c = [\vec{W}_d]_c \in \mathbb{R}^d$. Hence, to assign a name to the SAE neuron c , we choose the word s_c corresponding to the clos-

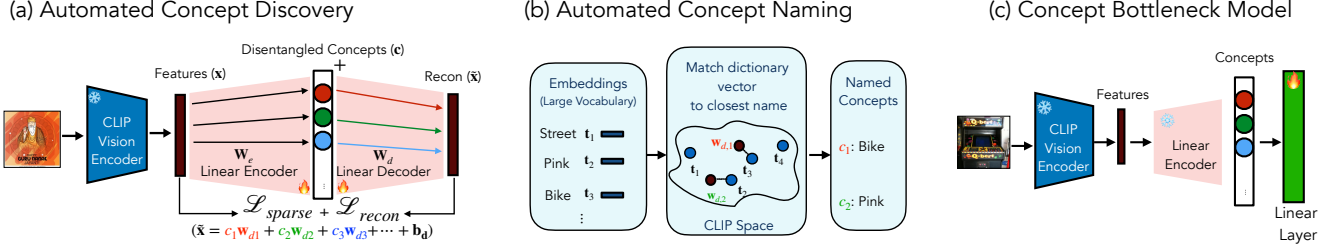


Fig. 1. **Overview.** Our approach consists of three steps (Sec. 2). (a) We train a sparse autoencoder on a large dataset (e.g., CC3M) to extract disentangled concepts from a CLIP vision backbone. (b) We use a large concept set of texts $\mathcal{V}$ to name each extracted concept. (c) We use the extracted and named concepts and train linear classifiers to construct inherently interpretable concept bottleneck models.

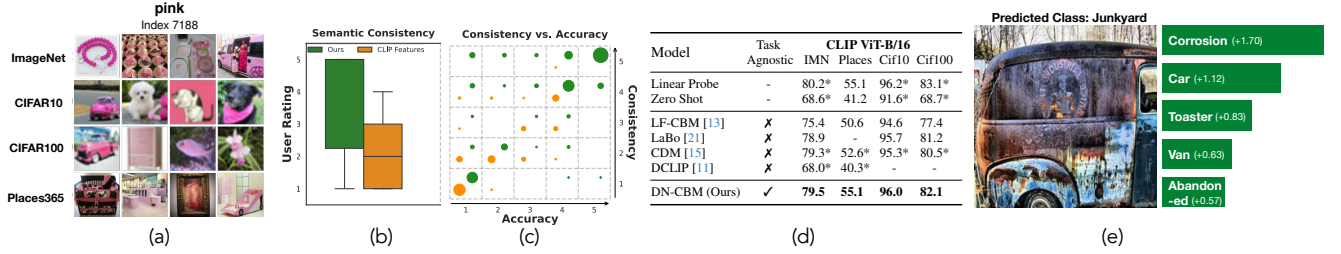


Fig. 2. Evaluation of our concept discovery and naming method (a-c); interpretability and performance of our DN-CBM (d-e).

est text embedding in $\mathcal{E}$: $s_c = \arg \min_{t \in \mathcal{E}} [\cos(\vec{w}_c, t)]$. We find our approach assigns meaningful and appropriate names to the SAE neurons (Fig. 2a-c).

Constructing the CBM (Fig. 1c): For each probe dataset, we now train a linear transformation [12, 17, 31] on the SAE’s frozen latents, yielding our CBM: $C(\vec{x}_i) = (h \circ (\phi \circ f) \circ \mathcal{I})(\vec{x}_i)$, where h is the linear classifier, $\phi \circ f$ is the SAE encoder, $\mathcal{I}$ is the CLIP vision encoder and $\vec{x}_i$ denotes the image. We use cross-entropy loss and an L_1 sparsity loss on the probe weights similar to [12, 17, 31].

3. Experimental Evaluation

Here we evaluate effectiveness of our concept discovery method, naming scheme, and performance of the CBM.

Task Agnostic SAEs (Fig. 2a): We visualize the top activating images for the concept #7188, which exhibit objects of pink color despite coming from very different datasets, and the automatically chosen name ‘Pink’ accurately reflects the common feature in the images. For more examples, refer to Sec. A.

Naming and Semantic Consistency of Concepts (Fig. 2b-c): We perform a user study to assess the concept consistency and naming accuracy quantitatively. We extract the top activating images for each concept from three datasets (ImageNet [6], Places365 [32], CC3M [28]), and for each concept, we ask two questions: (1) semantic consistency: if the top activating images map to some human interpretable concept, and (2) how accurate the assigned name is, if so. To evaluate if our SAE yields more disentangled concepts, we compare with CLIP neurons named using CLIP-Dissect

[16] as a baseline. Fig. 2c shows that semantic consistency of SAE neurons is higher than CLIP neurons, and Fig. 2d shows that our assigned names score highly in terms of accurately representing the concept (top right) as compared to the baseline. for more details about the user study, refer to Sec. F.

CBM Performance (Fig. 2d): DN-CBM is highly performant across datasets and backbones despite being task-agnostic, showing the generality of our approach. We get a high gain with Places365 (i.e. 52.58→55.11 pp.), which contains a variety of coarse objects, which are likely more well-represented in our concept space trained on CC3M.

Explanations of individual decisions (Fig. 2e): For an image, we show the most contributing concepts for the model prediction along with their contribution strengths. Here, the top concepts like ‘corrosion’, ‘car’ and ‘toaster’ are class-relevant, appropriate for the class ‘Junkyard’, thus aiding interpretability. For more examples, see Sec. C.

Additional results: See Sec. A for examples of concepts and names, Sec. B for impact of vocabulary on naming, Sec. C & Sec. D for local & global explanations, Sec. E for semantic consistency of SAE latents, Sec. F for user study details, and Sec. G for application of interventions.

4. Conclusion

We propose a novel approach to discover and name concepts known to a model. Our evaluations show that it is effective at discovering semantically consistent concepts with appropriate names and yields performant CBMs across datasets.

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